A Simulation-Based Multi-Criteria Charging Station Recommendation Framework for Electric Vehicles Using SUMO and TraCI

***Abstract*—The increasing popularity of electric three-wheelers in Bangladesh has caused stress on urban power infrastructure. This study suggests an EV Charging Station Recommendation System based on the implementation of Eclipse SUMO & TraCI. Using Eclipse SUMO & TraCI, a realistic scenario in the Khulna city with the use of heterogeneous vehicles is considered for the proposed multi-criteria decision making model for providing the recommendations of the stations based on SoC, occupancy, and load information. An experiment was conducted which shows a promising recommendation rate of 90.1%, whereas the recommendation rate of only considering nearest stations is merely 66%. The simulation showed a peak of 1.37 MW, which resulted in the collection of 33,721 data points for further analysis.**

***Index Terms*—Electric Vehicles, Easy Bike, E-Rickshaw, Charging Recommendation, SUMO, TraCI, Smart Grid, Load Balancing, Bangladesh**

I. Introduction

The global adoption of electric vehicles (EVs) has accelerated as a strategy to prevent greenhouse gas (GHG) emissions from the transportation sector, which is responsible for approximately 38% of total global GHG production [1]. In Bangladesh, this change has taken an individual form: instead of personal passenger cars, battery-powered three-wheeled vehicles – locally known as Easy Bikes and E-Rickshaws – have become the main intercity transportation medium [9].

Khulna, the third largest city in Bangladesh and an important industrial hub, is a case in point. Thousands of electric three-wheelers ply the streets of Khulna every day, providing affordable, low-emission urban mobility and supporting the livelihoods of a significant driver population [2]. According to Khulna's traffic department, there are 30,000+ battery-driven easy bikes in the city, with Khulna City Corporation (KCC) having licensed 8,000 and processing approvals for another 2,000 [2].

In spite of these facilities, EV charging in Khulna remains almost completely imbalanced. Drivers select charging stations based only on proximity and familiarity, which results in intense demand at popular stations during morning and evening rush hours. Transformers serving to those overloaded nodes - experience thermal stress, voltage drops, and failures. This results repair costs and disruption of driver service [14]. In the meantime peripheral stations remain continuously unused, which highlights the infrastructure investment. Electric Vehicle load on the electricity grid is projected to reach 780 TWh by 2030 [3], [4] globally. It makes intelligent charging coordination an urgent operational priority [16].

Previous work on EV charging management has mainly addressed advanced-economic contexts where vehicles are freely routable [5], [6] and have the advantage of inter-vehicle navigation. There are 2 limitations in the context of Khulna city which are absent in this literature : (i) Easy Bikes and Electric Rickshaws and Electric Vans follow specific corridor routes, as SUMO - the simulator used enforces one-way lane assignment; (ii) charging infrastructure is scattered and heterogeneous in terms of capacity, power rating and geographical distribution [15]. This is a purpose-built simulation framework that realistically models the Khulna fleet and designs around it rather than ignoring routing limitations.

The paper represents a SUMO+TraCI simulation framework that models a heterogeneous fleet of continuous 100 vehicles(Easy Bikes, Electric Rickshaws, Electric Vans) on 8 road-routes [10], [11]. It monitors real time battery SOC and also issues per vehicle charging station recommendations through a multi criteria scoring models. This framework distinguishes between potential routing – when a directed network path exists. It also suggests only output when directional limitations prevent routing. In comparison with nearest only baseline, significant improvements are seen in potential recommendation rates, congestion avoidance and genius handling of high priority charging events. The contributions of this work are: (i) a calibrated SUMO simulation of Khulna city's heterogeneous Electric Vehicle fleet with physics-based battery modeling on 3 vehicle classes; (ii) a multi-criteria recommendation algorithm that clearly maintains SUMO directed routing; (iii) comparative evaluation with nearest only baseline using four performance metrics; (iv) a 33,721 recorded vehicle centric dataset.

II. Related Work

Recent studies on planning infrastructure for EVs have relied on multi-objective optimization and MCDM techniques such as AHP and TOPSIS for station location and demand management [1], [2], [5], [6]. Real-time routing and AI-driven selection strategies such as reinforcement learning were explored via simulation environments such as SUMO to address grid loading problems and minimize traffic [7], [10], [12], [17]. The significance of feature engineering for forecasting was pointed out by Siddiqui et al. [8]. Still, there is a need to fill the research gap related to corridor-based fleets such as South Asian Easy Bikes through advising, keeping simulation consistency while accounting for different capacities of the batteries used [15]. This paper aims to fill those gaps with a simulated SUMO environment and multi-criteria algorithm.

III. System Architecture

Here the proposed structure consists of five components: the Road Network and SUMO Traffic Simulation Engine, Heterogeneous EV Fleet, Charging Station Infrastructure, the

TraCI Communication Layer with the Python based Charging Recommendation Engine and the Data Logging Module.

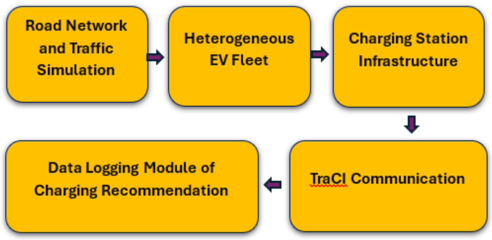
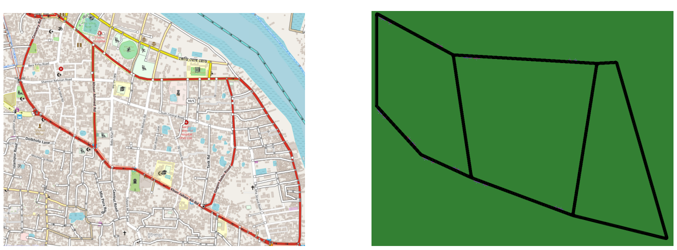


Fig. 1: System Architecture of the Proposed EV Charging Station Recommendation.

A. Road Network and Traffic Simulation

The simulated network creates a representative model of a part of Khulna city. It consists of 11 directed edges(E0-E10), bidirectional 8 designated routes(4 – forward, 4 - reverse). Traffic Light System controls the intersection model with realistic phase timing. The simulation runs from t=0 to t=600s with a 0.1s step resolution, with data collected every integer second. Time-to-teleport is disabled to prevent artificial vehicle movement or routing. Lateral resolution is set to 0.8m to properly model the narrow lanes of Khulna city, as shown in Fig. 2.



(a) (b)

Fig. 2: (a) A particular region's map of Khulna city from Open Street Map. (b) Simulated road (custom) network of Khulna city in SUMO, consisting of 11 directed edges (E0–E10) with 8 bidirectional routes and traffic-light-controlled intersections (red dots indicate junction nodes).

B. Heterogeneous EV Fleet

To model Khulna's real Electric Vehicle ecosystem, three vehicle classes with different battery configured: Easy Bikes(34 types), Electric Rickshaws(33 types), Electric Vans(33 types). Total 100 continuous flows across the network. From 2 to 81 vehicles per hour-launce vehicle through the 600 seconds simulation. Each vehicle has a SUMO battery device with specific values: 345 kg vehicle mass, 0.760 air drag coefficient, 0.0155 rolling drag, 81% propulsion efficiency, and 64% recovery efficiency [20]. Table I summarizes the three vehicle fleet classes. Significantly, the easy bikes carry 12,000 Wh batteries, whereas the e-rickshaws and e-vans are equipped with smaller batteries with 5,760–7,200 Wh batteries. This scenario reflectes the actual mixed-fleet composition deployed in Khulna. During simulation vehicles enter with different initial SoC levels(12.9% - 43.2%), that reflects realistic field conditions.

TABLE I: Heterogeneous EV Fleet Configuration

Vehicle Type	Count	Battery (Wh)	Max Speed (m/s)
Easy Bike	34	12,000	8.33
E-Rickshaw	33	5,760–7,200	6.94
E-Van	33	5,760–7,200	5.56

C. Charging Station Infrastructure

10 charging stations are implanted as simulation parking areas over 2-way, 5-road corridors. Each of them has separate charging sub areas and waiting sub areas. The charging sub areas have SUMO charging device parameters like-charging power, efficiency of charging. They enable local SUMO battery rechargeable. Table II explains detailed configuration. Here the total system capacity is 62 charging slots and 49-waiting slots. Station power rating range is : 25-50 kW with 95% efficiency.

TABLE II: Charging Station Configuration

Station	Chg. Slots	Wait. Slots	Power (kW)	Eff.
pa_1 (fwd)	4	3	25	95%
pa_1_rev	4	3	25	95%
pa_3 (fwd)	5	4	30	95%
pa_3_rev	8	6	40	95%
pa_5 (fwd)	4	3	25	95%
pa_5_rev	4	3	25	95%
pa_7 (fwd)	6	5	35	95%
pa_7_rev	6	5	35	95%
pa_8 (fwd)	5	3	50	95%
pa_8_rev	10	6	50	95%
Total	**62**	**49**	25–50	95%

D. TraCI Communication Layer

TraCI gives a socket based API. It interacts with SUMO and Python. In every integer second, the controller gathers battery parameters (`device.battery.chargeLevel`, `device.battery.capacity`, `device.battery.totalEnergyConsumed`), vehicle location, speed, current road ID, lane ID and stop status. It uses `traci.simulation.findRoute()` to evaluate topological reachability. This is an important step for directional constraints—and `traci.lane.getShape()` to calculate Euclidean station distances. Where a route change is possible and the vehicle's SoC drops below the threshold value of 30% warning. The controller reroutes the vehicle to its assigned parking area(used as charging area here) by calling `traci.vehicle.setRoute()` and `traci.vehicle.setParkingAreaStop()`.

call. That maintains orderly slot throughput without outside interference. Further, each recommendation event is logged as 28 attribute CSV record, generating 1,169 events and a vehicle-centric dataset with 33,721 timestep records.

IV. Charging Recommendation Model

A. Multi-Criteria Scoring Formulation

At simulation time t, for a vehicle v requiring charging, the proposed station-S^* reduces a composite score over the set of N=10 available stations:

$$S^* = \arg\min_i \left(w_1 D_i + w_2 Q_i + w_3 L_i\right) \tag{1}$$

Where, D_i = Euclidean distance from vehicle v to station s_i (m), Q_i = number of vehicles currently in the waiting queue at s_i, and L_i = instant charging power load at s_i (kW). Weighting factors w_1 = 0.5, w_2 = 0.3, w_3 = 0.2 were set to prioritise proximity while moderating congestion and load effects. Rankings are calculated using Euclidean distance for computational efficiency at every 1 second polling interval. The top two candidates (closest and second-closest). Then go through the decision logic below.

B. Priority classification of recommendations based on SoC of the vehicle

For every recommendation event generated by the system, the SoC of the vehicle is considered to assign one of the 3 priority levels – CRITICAL (when SoC $<$ 20%), HIGH (20% $\leq$ SoC $<$ 30%) and MEDIUM (30% $\leq$ SoC $<$ 50%). Assigning priority level to every recommendation event plays an important role for the following reasons:

- **CRITICAL** (SoC $<$ 20%): Immediate attention is needed since there is a chance of battery exhaustion during the journey.
- **HIGH** (20% $\leq$ SoC $<$ 30%): Urgent attention is required since it needs to charge at the next possible station.
- **MEDIUM** (30% $\leq$ SoC $<$ 50%): Proactive recommendation since the vehicle will be able to arrive at the station comfortably.

C. Directional Constraint and Advisory Mode

A notable innovation of this framework is SUMO's explicit handling of directional routing constraints. In Khulna, easy bikes and e-rickshaws operate on specific corridor routes. SUMO enforces one-way lane assignments and cannot teleport vehicles to the opposite end. As a result, the framework operates in two modes:

Rerouting Mode: When, traci.simulation.findRoute() - returns a valid directed path from the vehicle's current end to the target station end, with a positive route length, the framework calls `setRoute()` and `setParkingAreaStop()` to redirect the vehicle. A return route to the vehicle's original destination is added if possible.

Advisory Mode: When no valid guided path exists, because of the target station is at an unreachable distance from the vehicle's recent running-structure, the framework logs the best scoring station as suggested recommendation. It preserves simulation integrity and produces useful data for real-world notification systems.

D. Four-Rule Decision Logic

Given the nearest station s_1 and second nearest station s_2. The recommendation algorithm applies the following rules sequentially:

1) **Rule 1**: If s_1 has a free charging slot $\rightarrow$ recommend s_1 (Rerouting Mode if path exists).
2) **Rule 2**: If s_1 is full for charging but s_2 has a free charging slot $\rightarrow$ recommend s_2.
3) **Rule 3**: If s_2 also lacks charging slots but s_2 has waiting capacity $\rightarrow$ recommend s_2 (waiting queue).
4) **Rule 4**: If both s_1 and s_2 are fully saturated $\rightarrow$ recommend s_1 (Advisory Mode, flag for retry).

A 30 second cooldown per vehicle prevents recommendation changes. Queue management uses FIFO propagation:- when a charging vehicle leaves, the next waiting vehicle will automatically promoted to a charging slot via `traci.vehicle.resume()` and new `setParkingAreaStop()` call. That maintains orderly slot throughput without outside interference.

V. Simulation Setup and Parameters

Table III summarizes the complete simulation parameter set. `Test1.sumocfg` (SUMO configuration) specifies the road network (`Test1_with_TLS.net.xml`), vehicle flows (`Test1.rou.xml`) and charging infrastructure (`Test1.chargingstations.add.xml`). Battery output, charging station output, trip information and summary statistics are collected via SUMO's native output facility. The Python controller (`charging_recommendation`) connects through TraCI, iterates in 1-second integer steps, and exports recommendations datasets to timestamped CSV files after simulation completion at t = 600 seconds.

TABLE III: Key Simulation Parameters

Parameter	Value
Simulator	SUMO (Eclipse)
Interface	TraCI (Python 3.x)
Simulation duration	600 s
Time step resolution	0.1 s
Number of vehicle flows	100
Vehicle classes	Easy Bike, E-Rickshaw, E-Van
Number of routes	8 (4 fwd + 4 rev)
Network edges	11 directed
Charging stations	10 bidirectional
Total charging slots	62
Total waiting slots	49
Charging power range	25–50 kW
Charging efficiency	95% [21]
SoC trigger threshold	50%
CRITICAL priority	SoC $<$ 20%
HIGH priority	20% $\leq$ SoC $<$ 30%
Recommendation cooldown	30 s
Scoring weights (w_1, w_2, w_3)	0.5, 0.3, 0.2

VI. Results and Discussion

A. System Load and Charging Statistics

The simulation ended at 600 s. It generates 33,721 vehicle centric records(one vehicle per second) and 1,169 charging recommendation events. The system recorded a peak instant charging power of 1,370 kW with 40 vehicles charging concurrently. That reflects near saturation of the 62 available charging areas during peak demand time. Across the simulation the average charging power was 963.7 kW, the total energy consumption for charging was 127.5 kWh. These load metrics highlights that there is an important charging demand on the network. It makes intelligent station selection essential to prevent local transformer overload.

B. Recommendation Volume and Priority Distribution

Table IV shows the priority level breakdown of the 1,169 recommendation events. HIGH-priority events (SoC 20%–30%) dominated at 42.7%, followed by MEDIUM (34.6%) and CRITICAL (22.7%). The average SoC at the time of recommendation was 27.7% (σ = 8.5%), with a minimum of 12.0% and a maximum of 46.5%. It ensures that the 50% trigger activates recommendations with sufficient battery margin. The average Euclidean distance to the nearest station at the time of recommendation was 84.3 m (σ = 62.0 m, min = 0.5 m, max = 256.6 m). It is indicating adequate station density for the modelled fleet.

TABLE IV: Recommendation Priority Distribution

Priority	SoC Range	Events	Pct.
CRITICAL	<20%	265	22.7%
HIGH	20%–30%	499	42.7%
MEDIUM	30%–50%	405	34.6%
Total	—	**1,169**	**100%**

C. Recommendation Outcome Analysis

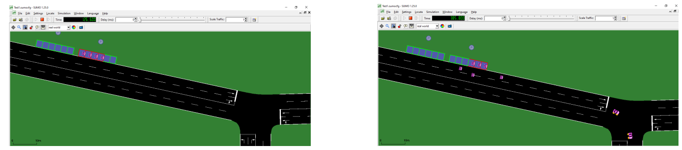
Table V shows the distribution of the results of recommendations. The main result — when charging slots were available, 59.0% (689 events)—was routing to the nearest station. In 24.1% of cases (282 events), the nearest station was fully occupied and vehicles were successfully directed to the second nearest station with available capacity. It is demonstrating genuine load diversion. Where both stations were saturated, Waiting slot overflow management handled 7.7% (90 events), while 9.2% (108 events) triggered Advisory Mode.

TABLE V: Recommendation Outcome Analysis

Outcome	Events	Pct.
Nearest station (slot available)	689	59.0%
2nd nearest (nearest full)	282	24.1%
Waiting slot overflow	90	7.7%
Advisory Mode (both saturated)	108	9.2%
Total	**1,169**	**100%**

D. Station Utilisation Distribution

Station pa_8 had the greatest proportion of recommendations (32.4%), which is attributed to its larger size and location in a central point. All the ten stations had recommendations, indicating that the multi-criteria score-based approach [7], [12] is capable of distributing demand well. Figure 3 shows dynamics in the simulation in real time of station pa_2, whereby part (a) shows multiple vehicle arrivals at t=126.5 s, while part (b) illustrates the vehicle departures at t=385.0 s.



(a) (b)

Fig. 3: Snapshots of vehicle (a) entry and (b) exit at charging station pa_2 corridors.

E. Comparative Evaluation: Proposed vs. Nearest-Only Baseline

To measure the benefit of the structure, we define a nearest-only baseline that always recommends the Euclidean nearest station without considering occupancy, queue, or load. Table VI shows 4-performance metrics computed from the 1,169-event dataset.

TABLE VI: Performance Comparison: Proposed vs. Nearest-Only Baseline

Metric	Proposed	Baseline
Feasible Rec. Rate (FRR)	90.1%	66.0%
Congestion Avoid. Rate (CAR)	37.1%	0.0%
Hi-Priority Smart Dec. (HPSDR)	27.9%	0.0%
Avg. Distance (m)	100.1	84.3

The following subsections explain each metric in depth - its definition, calculation method, numerical values and significance in this work.

1) Feasible Recommendation Rate (FRR): FRR is the fraction of all recommendation events , where at least one free charging slot was available at the station recommended by the algorithm, at the time of the recommendation. A recommendation is “feasible”, if a vehicle is sent to the station can start charging immediately without waiting for a slot to become free.

Calculation Formula:

$$\mathrm{FRR} = \frac{N_{\text{rec. station}\geq 1\text{ free slot}}}{N_{\text{total events}}} \times 100\% \tag{2}$$

For the proposed framework, the feasible events are those resolved by Rule 1 (689 events, nearest with free slot) plus Rule 2 (282 events, 2nd nearest with free slot) = 971 feasible events out of 1,169 total.

$$\text{FRR (Proposed)} = \frac{971}{1{,}169} \times 100\% \approx 90.1\% \tag{3}$$

For the baseline, the nearest station is only possible int the 771 events (66.0% of 1,169) where there was a free slot at the nearest station , regardless of diversion. The baseline doesn’t attempt to check or redirect to a better option.

Interpretation: An FRR of 90.1% means that 90 out of every 100 recommendation events, are taken to a station where it can immediately plug in. The 24.0 percentage-point improvement over the baseline quantifies the concrete benefit of occupancy-aware selection: vehicles are far less likely to arrive at a full station, waste time waiting, or drain their already-depleted battery in further searching.

2) Congestion Avoidance Rate (CAR): CAR computes the proportion of all recommendation events , where the proposed framework successfully diverted a vehicle away from a congested (fully occupied) nearest station to an alternative station with available capacity. It directly quantifies the load-balancing effectiveness of the recommendation engine.

Calculation Formula:

$$\text{CAR} = \frac{N_{\text{nearest full \& alt. with free slots}}}{N_{\text{total events}}} \times 100\% \tag{4}$$

This corresponds exactly to Rule 2 events: 282 cases in which the nearest station was fully occupied but the second-nearest had capacity. The baseline, which blindly recommends the nearest station, contributes 0 such diversions.

$$\text{CAR (Proposed)} = \frac{282}{1{,}169} \times 100\% = 37.1\% \tag{5}$$

$$\text{CAR (Baseline)} = 0.0\% \tag{6}$$

Interpretation: A CAR of 37.1% means that more than one in three charging events, which would have overloaded a popular station are instead redistributed to a less congested alternative. In grid terms. This translates directly to reduced transformer peak loading, lower risk of thermal stress, and more uniform utilisation of the installed charging infrastructure. The baseline's CAR = 0% confirms that proximity-only dispatching cannot achieve any congestion relief by design.

3) High-Priority Smart Decision Rate (HPSDR): HPSDR measures the proportion of CRITICAL (SoC $< 20\%$) and HIGH ($20\% \leq$ SoC $< 30\%$) priority recommendation events — those carrying the highest battery urgency — in which the framework successfully moved the vehicle to the second-nearest station because the nearest station lacked available charging slots. It captures intelligent decision-making precisely when the stakes are highest: when battery depletion in transit is a realistic risk.

Calculation Formula:

$$\text{HPSDR} = \frac{N_{\text{CRIT+HIGH: chose 2nd over congested 1st}}}{N_{\text{total events}}} \times 100\% \tag{7}$$

CRITICAL events: 265; HIGH events: 499; combined = 764 high-urgency events out of 1,169 total (65.4%). Of these, the framework successfully diverted to the second-nearest station (Rule 2 applied during CRITICAL/HIGH events) in 27.9% of all events = 326 events.

$$\text{HPSDR (Proposed)} = \frac{326}{1{,}169} \times 100\% = 27.9\% \tag{8}$$

$$\text{HPSDR (Baseline)} = 0.0\% \tag{9}$$

Interpretation: A value of HPSDR of 27.9% shows that the framework does not simply assume the nearest-station policy even under high levels of battery urgency but, rather, reasons about availability and allocates the most critically affected vehicles to the nearest possible station. A 27.9% difference between our framework and the baseline is highly relevant in the case of Khulna, where being out of battery midway on a journey means that the Easy Bike or E-Rickshaw gets stuck and causes traffic as well as income loss for the operator. Baseline HPSDR is equal to 0% since there is no way the baseline framework can diverge from the nearest station due to the lack of an urgency level parameter.

4) Average Distance to Recommended Station: This measure gives the average straight-line distance (in meters) between a vehicle's location at the time of recommendation and the recommended station. It acts as the cost factor to be balanced against the benefit gained from congestion avoidance since routing away from the nearest station increases travel distance.

Calculation Formula:

$$\overline{d} = \frac{1}{N} \sum_{i=1}^{N} d_{\text{Eucl.}}\big(\text{pos}_i,\ s_{i,\text{rec}}\big) \tag{10}$$

where N = 1,169 total recommendation events, pos_i is the vehicle's geographic coordinates at event i and $s_{i,\text{rec}}$ is the centre coordinates of the recommended station. The Euclidean distance is measured using `traci.lane.getShape()` coordinate data polled at each integer second.

$$\overline{d}_{\text{Baseline}} = 84.3\text{ m} \quad \text{(always the nearest station by definition)} \tag{11}$$

$$\overline{d}_{\text{Proposed}} = 100.1\text{ m} \quad \text{(can be the 2nd nearest)} \tag{12}$$

Interpretation: The suggested framework asks vehicles to travel, on average, 15.8 m further than the nearest station. This is an accepted cost to avoide crowded stations. To put this tranasaction in to context, consider that the average SoC at recommendation time is 27.7% and the minimum battery capacity is 5,760 Wh. Even at an average propulsion power of $\sim$1.5 kW and a speed of 10 m/s. The additional 15.8 m adds less than 2 seconds of travel time and consumes a negligible energy increment well within the available battery margin. The trade-off is therefore, operationally inconsequential while the gains in FRR (+24.0 pp), CAR (+37.1 pp), and HPSDR (+27.9 pp) are substantial.

5) Summary Comparison and Interpretation: The table VI combines all four metrics. The proposed multi-criteria framework outperforms the closest only baseline on every effectiveness dimension while enduring a small distance penalty. The combined information is as follows :

FRR (90.1% vs 66.0%): 9 in 10 recommended stations have free slots immediately versus only 2 in 3 under naive routing. This reduces wasteful station to station re-querying by vehicles already at low SoC.

CAR (37.1% vs 0.0%): The framework actively load-balances by redirecting more than a third of all events away from congested hubs to underutilised alternatives, reducing transformer peak stress and prolonging infrastructure lifetime.

HPSDR (27.9% vs 0.0%): Even the most critically battery-depleted vehicles receive improved routing — the framework applies its highest priority triage precisely, while failures are most costly in the real world.

Avg. Distance (+15.8 m): The marginal additional travel introduced by smarter routing is 15.8 m — less than 2 vehicle lengths — and is operationally negligible relative to the energy and time benefits.

F. Limitations and Future Work

A simple network is included in the 600-second simulation window. Full scale deployment requires GPS trace calibrated OD matrices and different charging profiles from field surveys.

Advisory Mode (9.2% of occurrences) can be reduced by increasing station density or adding a 3rd nearest candidate. Dynamic electricity, pricing and renewable energy at station nodes will be integrated in future work.

VII. Conclusion

This paper shows a SUMO+TraCI simulation-based charging station recommendation framework for heterogeneous EV fleet in Khulna city. A 100 flow simulation through 3- vehicle classes with varying battery capacities (12,000 Wh for Easy Bikes; 5,760–7,200 Wh for E-Rickshaws and E-Vans), eight routes and ten bidirectional stations generated 1,169 recommendation events over 600 seconds. Also it depicts the record of peak charging power of 1,370 kW with 40 vehicles simultaneously charging and consuming 127.5 kWh total energy.

The main contributions are: (1) a physics-calibrated SUMO fleet model with different battery configurations across three vehicle classes; (2) a multi-criteria recommendation algorithm with Advisory duality; (3) comparative evaluation demonstrating +24.0 pp potential Recommendation Rate, +37.1 pp Congestion Avoidance Rate, and +27.9 pp High-Priority Smart Decision Rate versus a nearest-only baseline; and (4) two complementary datasets (33,721 vehicle-centric records + 1,169 recommendation events).

Future work will be expanded to include GPS trace calibration of the full road graph of Khulna, dynamic pricing, model renewable energy at stations and evaluate demand-response schedules consistent with Bangladesh's National Energy Policy.

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